1. Executive result and delivery boundary

The Freight Invoice Itemizer turns carrier dispute descriptions into categorized charge claims, stated or calculated amounts, and the exact supporting words. The current-data implementation, evaluation, exception review and assisted labeling are complete for the supplied export. This is a bounded delivery of a reviewable extraction system, not a certification of payable freight expenses. Accountants retain payment and allocation decisions.

The independent OpenAI experiment processed all 729 distinct nonblank notes and all 1,347 source rows. On the existing 200-note reference, 189 of 197 answered notes matched categories and amounts: 95.94%. Counting abstentions as failures gives 189/200, or 94.5%. The original references were model-drafted; the user attested review of all 200 separately. Neither figure establishes independently double-labeled accuracy across the complete export. The strict result also does not meet a 95% strict-yield requirement.

The client demo took place on November 5. Its transcript clarified that repeated source rows come from database joins, and that the requested output is unique charge evidence at shipment level, keyed by SAP BOL plus description. The offline export preserves every source row while reducing repeated join records to 1,284 shipment/description groups. Finance performs downstream invoice allocation. Generic balances and already-paid amounts are context rather than new expense spend.

The final review version covers 245 groups: every originally flagged group, related descriptions and one subsequent user-confirmed interpretation. There are 209 completed extraction reviews and 36 groups requiring unavailable evidence. The analytical proposal holds 68 claim items, consolidates 49 repeated fee entries, and moves 81 entries to balance context. It contains 1,249 expense candidates and 94 context items; some candidates outside the focused review retain their original model status. These are record counts, not approved-spend totals or a new accuracy result.

The delivery preserves raw predictions, historical scores and human decisions. No unresolved amount, rate, date, charge type or SAP BOL was invented to close an exception. The work is ready for current-data handoff with disclosed limitations. Original requirements for independent human review, client acceptance and fresh-export validation remain unsatisfied unless their owners explicitly accept a revised scope. This report does not record such a revision or sign-off.

2. Problem, source data and analytical grain

The source export describes differences between estimated and carrier-billed freight. Its free-text notes may name accessorial charges, rate corrections, taxes, payments, balances, weights or dates. A number occurring in a note is not automatically a charge. The extraction task must identify what the carrier actually states while retaining evidence and uncertainty. Whether the claim is valid under a contract or should be paid is a separate task.

The supplied workbook contains 1,347 nonempty source rows, 729 distinct normalized nonblank descriptions, and 18 carriers. The dataset identity is SHA-256 e08da6c7fd6e03036d537a13ed0a52748ef81fc5cedea0df7f3ec037bc43c0e5. Source notes and identifiers remain private. No raw workbook, source text, predictions, credentials or review ledger is included in the public repository.

There are three useful grains. The model operates on distinct descriptions so identical normalized notes can reuse an answer. The source-row layer preserves the original records and row outcomes. The shipment-review layer groups SAP BOL and description, retaining the source-row, order/invoice and shipment references needed to audit each extracted item. These grains have different item counts; a distinct-note item count must not be confused with its shipment occurrences.

SAP BOL comes directly from column E. It is not replaced by SHIPMENT_GID or the carrier invoice number. The source has 1,341 text BOL cells, six missing BOLs and 1,154 distinct nonmissing BOL values. Long identifiers and leading zeros are preserved as text. Missing BOL records remain separate review records rather than being grouped under one blank key.

Grouping produces 1,284 description groups and removes 63 repeated join-row occurrences from counting. A client-demo example with 24 source rows reduces to three charge-description groups while retaining all original references. Different descriptions can still represent repeated or revised charges. The review layer therefore flags overlapping categories and does not simply add every description into a financial total. Source dispute values remain available for reconciliation, but an extraction is never edited merely to equal them.

3. Extraction and evidence method

The implementation separates data loading, note grouping, model providers, structured-answer validation, source-row assignment and reporting. The taxonomy has 19 named categories plus other_unmapped. It covers purchased transportation, base-rate corrections, fuel, handling, detention, storage, port/drayage services, taxes and other common freight concepts. Definitions distinguish close concepts such as driver waiting versus equipment held for days.

Each extracted item records its category, amount, evidence quotation and character offsets, line type and derivation. A stated amount is distinct from a product such as days times daily rate. Date strings, BOL weights, mileage rates and invoice totals are not treated as extra charge items merely because they contain numbers. A missing quantity is not filled with zero, and arithmetic does not establish that the stated days were actually chargeable.

Validation checks enforce the answer schema, permitted categories, positive amounts, evidence spans and supported derivations. An exact quotation is an audit aid, not proof of correct interpretation. Likewise, zero invariant violations do not mean every charge was categorized correctly. A note that cannot be resolved can abstain or enter a review queue rather than acquire an invented answer. Text telling a carrier or accountant to add a fee is treated as source evidence of a claim, not as an instruction to the AI or approval to pay.

The offline shipment exporter consumes saved note results and the source workbook. It makes no model requests. An optional row-scoped overlay records specific reviewed decisions without mutating the original predictions. The overlay is bound to the source workbook hash and expected BOL/description fingerprint. Conflicting or partially scoped corrections on joined rows fail closed. The exporter also distinguishes real BOL values from missing-key markers.

Already-paid amounts and generic short-paid balances are exported as context. Overlapping categories across descriptions are flagged instead of automatically summed. The subsequent AI review supplies an explicit analytical proposal for known repetition and ambiguity patterns. That private downstream proposal is separate from the raw candidate CLI output. Review decisions are attributed to the actual source: user, delegated AI recommendation or unresolved evidence.

4. Experiment protocol, metrics and spending control

Alejandro's original prototype, shared provisional references and the reconstructed local split were preserved as distinct experiment records. Their membership and freeze history cannot be combined merely because they contain similar notes. The local readiness correction pins prompt examples to TRAIN before creating a new split; already locked splits are not silently changed. Historical results remain attached to the code, data, prompt, taxonomy and reference identities under which they were produced.

The separate OpenAI candidate used the pinned gpt-5.4-mini-2026-03-17 snapshot, medium reasoning, a 16,000-token completion limit, standard service tier and store=false. The original source checkout remains separate so completed Gemini and OpenAI protocols can retain their frozen source identities. New exporter work lives on a separate development branch; a changed package hash is not passed off as the original completed inference experiment.

The main benchmark comparison matches categories and amounts within one cent. Answered-note agreement excludes abstentions from its denominator; strict yield includes every reference note. Row coverage includes covered discrepancies, which means a row can count as covered while still requiring manual attention. Dollar coverage is not dollar correctness or the percentage of dollars approved for payment. Weighted figures depend on the population and weighting rule and must be identified separately from unweighted note agreement.

The user authorized a maximum of $5 total for OpenAI evaluation. A persistent shared ledger reserves a conservative maximum before each paid request and settles it against metered usage. Unknown costs retain their reservations. The completed experiment used $1.05022950, including development checks and validation retries; no unsettled reservations remained. The cap does not cover unrelated activity in the same account. Review, labeling and shipment exports after that run used saved results and made no additional model/API requests.

The 50-note assisted-label pass was finalized and hashed before reading its corresponding saved predictions in that pass. However, inference on the old export had already occurred and the assistant had project context. This is not a new independent blind evaluation. Human substitutions and policy corrections are not credited as an improvement to the original model benchmark.

5. Completed evaluation and practical limits

The benchmark returned 197 answers and three abstentions. It matched 189 notes exactly on category and amount. Answered-note agreement was 95.94%, with a 95% Wilson interval of 92.19%-97.93%. Strict yield was 94.5%. Reference-stratum row-weighted agreement was 96.41%, with a weighted bootstrap interval of 93.15%-98.80%. These estimates describe agreement with the existing reference, not a statistically guaranteed floor for future production accuracy.

The benchmark's eight answered disagreements included category and amount errors, a missing item and an extra item. Performance was weaker on difficult subsets: large-dispute row-weighted agreement was 85.29%, and unseen-format agreement was 48/54, or 88.89%. The overall benchmark exceeds the user's 90% target, but those subgroups do not establish a 90% floor. A 95% strict-yield acceptance target is not met by the 94.5% strict result.

The full sweep completed 729 distinct notes and 1,347 source rows. Row outcomes were 1,084 verified, 12 no-charge, 240 discrepancy and 11 review. Coverage was 1,336/1,347 rows, or 99.18%, and reported disputed-dollar coverage was 99.19%. Only 1,096 rows, or 81.37%, were accepted without reading under that run's row rules. Their disputed dollars represented 49.57%, illustrating that high overall coverage can coexist with substantial high-value exception exposure.

There were 879 verifiable distinct-note items in the full sweep and no invariant violations. All benchmark predictions were identical when reused in the sweep. The complete run establishes processing coverage, not independently verified accuracy on the 529 notes outside the existing 200-note reference. An all-scope weighted approximation is therefore not used as the headline accuracy result.

The later shipment and assisted-label reviews change the analytical treatment of some claims and balances without rewriting these historical measurements. In the 50-note assisted pack, 44 named-fee notes agree with saved expense outputs; six balance-context treatments differ under the reviewed policy. That comparison is neither an independent 88% accuracy estimate nor a new passing benchmark. It identifies policy differences that belong in the reviewed output and report.

6. Review, assisted labels and exception decisions

The first priority pass recorded 18 note-level cases using explicit user answers or delegated recommendations. It distinguished already-paid amounts from outstanding balances, preserved duplicate source rows, confirmed stated tax, hazmat and other fees, and deferred unsupported rates, categories and revisions. Subsequent source auditing retained invoice dossiers rather than fabricating payment histories or assuming row order established chronology.

At shipment grain, the original export flagged 189 groups. Review included those groups, related descriptions, 80 purchase-fee headings with only balance-due amounts, and a later user-confirmed balance interpretation. The final scope is 245 groups: 209 extraction reviews completed and 36 requiring external evidence. The unresolved groups are consolidated into 21 shipment/invoice evidence requests. No request was sent automatically and no reply is assumed.

The separate analytical proposal consolidates 49 repeated fee entries with the same reviewed BOL/invoice/category/amount across alternate descriptions. It preserves all original claims and canonical references in the audit. This is a review proposal, not a universal rule that every equal amount is the same fee. Two item ordinals in one note are not collapsed. Taxes tied to different named fees remain separate. Explicitly conflicting fee versions are held, not added together or assigned an unsupported latest version.

The 50-note labeling pack contains 45 expense-claim items across 44 notes and six balance-context notes. Exact evidence quotations were checked. The user made one explicit interpretation decision, recorded separately and propagated to the updated proposal. The other labels were authored by the AI assistant. The pack is complete as AI-assisted labeling; it is not evidence of two independent human readers, nor is one user answer approval of every label or client sign-off.

Because missing information is unavailable, the proposal excludes 68 held claim items rather than repeatedly asking the user to guess. Missing BOLs, ambiguous rate/day evidence, detention type and revised fee amounts remain disclosed. There are 1,249 proposed expense candidates and 94 context records. Candidate records outside the focused review retain a model-candidate status. No mixed currency financial grand total, validated invoice liability or automatic payment decision is produced.

7. Reproduction, verification and handoff

The repository contains the source, configuration, synthetic-data tests, report and rerun guide. Private inputs and outputs live under ignored data/. The existing-data shipment export can be reproduced offline from the source workbook, saved all-scope note results and optional source-bound overlay using python -m itemizer.shipment_review. The detailed command, input layout and expected counts are in deliverables/final_rerun_guide.md.

Saved inference should be audited in its matching frozen checkout. Resuming an old experiment from changed source can correctly fail protocol checks; it is not a reason to weaken those checks or copy a new hash over the old protocol. A new experiment needs its own identity and an explicit budget/authorization decision. Cached, paid and reviewed phases must remain distinguishable.

The private handoff has two layers. The raw shipment candidate export supplies 1,284 description groups and all source links. The later reviewed workbook adds the completed assisted labels, review decisions, proposed candidates, context, claim audit and source rows. The original raw export is not overwritten. The AI review and assisted-label analysis scripts and their input hashes are saved privately because their concrete decisions refer to confidential source cases. Repository collaborators need those artifacts through an approved private channel to reproduce the exact reviewed snapshot; Git alone is insufficient.

Verification covers join-row collapse, distinct shipment identities, missing BOL isolation, source import, stale/partial overlays, repeated item ordinals and balance treatment. Review fixes were tested with invented notes, including a balance item whose evidence quotes only its amount. Export checks read back all source BOL values as text, preserve leading-zero invoices, account for every claim item, check canonical references and confirm that user decisions reach the downstream output. Workbook tabs were rendered for visual inspection.

The final closeout runs the complete automated test suite and records its result in the repository handoff. Credentials, private notes and generated workbooks are excluded from the public push. No inference run or additional API spending is necessary for this verification. A successful technical test run establishes software behavior; it does not manufacture contractual acceptance evidence.

8. Limitations, acceptance and closure

The delivered version is bounded to the supplied export and recorded review decisions. Carrier descriptions express claims, not necessarily contractually valid charges. A convincing quotation, correct arithmetic or a reconciled source delta does not prove a service occurred or a fee was payable. The held items are excluded from the proposed expense candidates, but the evidence and unknown values remain available for a future reviewer. This is preferable to silently filling missing information or declaring every source row settled.

Three original acceptance conditions remain external. The scope calls for independent human labeling/review, client acceptance and validation on a genuinely new export. The existing assisted pack cannot be renamed to satisfy the first; the November 5 demo cannot automatically satisfy the second; repeated testing on the inspected old export cannot satisfy the third. An authorized client/course owner may agree a revised scope accepting these limitations, but no such agreement is invented or recorded by this closeout.

There is no pending freight-interpretation question for the user in the completed 50-note pack. The requested source judgments have been resolved or the missing facts explicitly marked unavailable. The remaining decision is whether the owners accept the current-data deliverable with its documented exclusions and AI-assisted provenance, or require the original validation milestones before formal acceptance. The final code/report handoff can proceed without claiming that those milestones occurred.

Historical evidence is summarized in deliverables/openai_evaluation_results.md, deliverables/shipment_review_handoff.md and deliverables/ai_review_2026-11-09.md. The private dataset, experiment protocols, case decisions, review ledgers and label-lock hashes support the aggregates reported here. The November 5 meeting record supports the shipment-level output requirement. The public rerun guide explains which private inputs a collaborator must obtain separately.

The result is a reproducible prototype with completed current-data processing, explicit evidence, a reviewed analytical proposal and a transparent exception boundary. The delivery does not promise that all future formats exceed 90%, claim a 95% strict pass, or authorize automated settlement. Future work should begin with real new data and independent labels, using a new experiment version rather than optimizing against the repeatedly inspected benchmark.